

Question number	Answer	Additional guidance	Mark
5(a)(i)	- tangent correctly drawn (1) - rate calculated (1) - rate value between 0.060 and 0.090 expressed to 3 dps max (1)	line touches the outside of the curve at 1 200 seconds e.g. $65 \div 880 = 0.073863636363$ $50 \div 720 = 0.069444444$ ECFs: $147 \div 1120 = 0.1386792$ $147.5 \div 1120 = 0.131696428571$ $148 \div 1120 = 0.13962$ $147 \div 1130 = 0.13725490$ $147.5 \div 1130 = 0.13053097$ $148 \div 1130 = 0.1383177$ Ecf values from mp 2 correctly rounded to a max of 3 dps Ecf value of 0.1225 / 0.122916 / 0.12333 correctly rounded to max 3 dps but do not accept recurring numbers	(3)

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5(a)(ii)	An explanation that includes three of the following points: - oxygen is produced by photolysis (1) - presence of PBF (on <i>C. pyrenoidosa</i>) produces {oxygen faster / more oxygen} (than <i>C. pyrenoidosa</i>) OR PBF (alone) produces {no / very little} oxygen (1) {no / very little} oxygen produced by PBF) as it is {only a molecule / cannot photosynthesise} (1) (more oxygen produced) because {more light absorbed / wider range of wavelengths absorbed / green light absorbed as well} by PBF (1)	ACCEPT presence of PBF means that light (intensity) is not rate limiting ACCEPT converse for algae alone producing less oxygen	(3)

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5(b)	An explanation that includes the following points: - because {faster light-dependent reaction / more ATP produced / more reduced NADP produced} (1) - more {GALP / glucose} produced {light-independent / Calvin cycle / carbon fixation} (by <i>C. pyrenoidosa</i>) with PBF) (1) - therefore (more) {large organic molecules / NPP / biomass} produced (1)	ACCEPT NADPH DO NOT ACCEPT NAD ACCEPT faster {light-independent / Calvin cycle / carbon fixation} to produce {GALP / glucose} (by <i>C. pyrenoidosa</i>) with PBF) ACCEPT named polymer e.g. starch, protein, cellulose	(3)